Sathvik Vempati

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Education

University of California, San Diego (UCSD) — La Jolla, CA
B.S. Mathematics–Computer Science | Junior | GPA: 4.00 — *Expected June 2028*

Experience

Software Engineering / Research Intern — June 2025 – Present
University of California, Berkeley – RadWatch — *Berkeley, CA*
- Built a live data pipeline for IoT radiation sensors (Geiger counters), processing 10k+ readings/sec in async Python.
- Cut lost sensor data from 8.2% to 0.05% at peak traffic using ring-buffered UDP sockets that absorb network bursts.
- Developed REST APIs and automated map-based radiation reports (GeoPandas), replacing manual analysis.

Research Assistant, Neuroscience & Software Engineering — Sept. 2025 – Present
Stanford University — *San Jose, CA*
- Architected the FastAPI backend of a real-time brain–computer interface detecting stress from EEG; cut latency 34%.
- Increased throughput of the real-time pipeline streaming 500 Hz EEG data by adding Redis caching layers.
- Sped up multi-channel EEG brainwave signal cleaning 42% by vectorizing filter pipelines in NumPy/SciPy.

Software / Data Science Intern — Dec. 2024 – May 2025
IPMD — *San Mateo, CA*
- Standardized PySpark ingestion pipelines for financial data feeds, cutting data-cleaning time by 28%.
- Made a deep learning model 5x faster (240 ms to 45 ms) using NVIDIA TensorRT compilation and quantization.
- Raised accuracy of an LSTM model forecasting stock indicators 14.5% via AdamW optimizer tuning (TensorFlow).

Software AI Trainer — Remote
Scale AI
- Evaluated and ranked LLM-generated code for correctness, efficiency, and readability to create training data (RLHF).
- Wrote programming prompts and reference solutions that test model reasoning on algorithmic and debugging tasks.
- Identified bugs and logic errors in model outputs and wrote clear rationales used to improve model responses.

Projects

ElasticSeer | *Python, FastAPI, Elasticsearch, Model Context Protocol (MCP), Docker, LLM Agents* — Oct. 2025
- Built an AI agent that reads system logs to pinpoint why software fails, cutting mean time to resolution (MTTR) 45%.
- Gave the LLM agent live access to logs through MCP servers, so it searches and interprets them on its own.
- Deployed custom log shippers in Docker containers, cutting infrastructure monitoring overhead by 18%.

Reflexiv-PR | *Python, FastAPI, Gemini, Docker, Daytona, Sentry, GitHub API*
- Built a self-healing AI agent that catches Sentry crash reports and reproduces each bug in an isolated Docker sandbox.
- Designed a Gemini prompt loop with custom Git parsing that reads stack traces, writes a fix, and reruns the tests.
- Goes from crash to a working GitHub pull request in under 64 seconds, with no human in the loop.

311.ai | *Strands Agents, Bedrock AgentCore, Amazon Nova, Redis, FastAPI, MCP* — Sept. 2026
- Built a team of AI agents that runs a city's 311 repair dispatch: a coordinator plus 3 specialist agents with 19 tools.
- Agents handle routine jobs end to end and escalate public-safety and budget calls to a human, freeing dispatchers.
- Live voice intake (Nova Sonic) and Redis vector search that merges duplicate reports of the same problem.
- Scored crews on skill, distance, track record, and workload to send the best team, with every reason logged.

Dense-Tune (Merged Open-Source PR) | *PyTorch, PerforatedAI, Weights & Biases, CNNs* — Jan. 2026
- Contributed an example merged into PerforatedAI's main repo applying dendritic optimization to a PyTorch CNN.
- Raised CIFAR-10 accuracy from 84.59% to 86.85% (14.7% fewer remaining errors) as the model auto-grew capacity.
- Built a Weights & Biases sweep over 7 hyperparameters so every experiment is tracked and reproducible.

Technical Skills

Languages: Python, SQL, C++, TypeScript, Bash, HTML/CSS
Frameworks & Infra: FastAPI, Docker, Git, CI/CD, GCP, React, Next.js, React Native, Node.js
Data & Systems: PostgreSQL, Redis, PySpark, Elasticsearch, Async Pipelines, REST APIs, NumPy, SciPy, Pandas
AI/ML & LLMs: NVIDIA TensorRT, PyTorch, TensorFlow, CNNs, Quantization, LLM Inference, Multi-Agent Systems, RLHF, RAG, LangChain, LlamaIndex, Vector DBs, MCP, Strands Agents, Amazon Bedrock

Hackathon Awards

CodeForge Labs | *2nd Place – Google Cloud Agent Hacks* — Oct. 2025
- Multi-agent platform where Claude Code, Gemini, and Browser-use agents test a live website, find usability problems, edit the code, and deploy fixes in real time (Next.js, Inngest, PostgreSQL).

311.ai | *Winner, Best Use of TokenRouter (PaleBlueDot AI) – UC Berkeley AI Hackathon 2026* — June 2026
- AI 311 hotline turning a resident's call into an assigned repair crew in under 60 s: Deepgram voice, multi-agent crew debate, Browserbase permit-filing agents, Redis dedup, Arize Phoenix tracing.

CharlieTourGuy | *2nd Place, Best Voice Agent – The AI Champion Ship* — Feb. 2026
- Voice AI tour guide that remembers context and location, pairing ElevenLabs voice, Google 3D Maps, and Vultr LLM inference for low-latency spoken answers (Hono.js, Raindrop).